

Write Algebraic Expressions

Lesson 6-3

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In $5t + 20$, t could be hours worked — if $t = 3$, the expression equals 35; if $t = 8$, it equals 60

Variable

A letter that stands for a number that is unknown or can change.

$3n + 7$ means 'triple a number, then add 7' — it has a variable (n), a coefficient (3), and a constant (7)

Algebraic Expression

A math phrase that has at least one letter.

In $8x$, the coefficient 8 means '8 groups of x ' — if $x = 3$, then $8x = 8 \times 3 = 24$

Coefficient

The number in front of a letter, like the 3 in $3x$.

In $3n + 7$, the 7 never changes no matter what n is — like a flat fee that stays the same

Constant

A number on its own that does not change.

$4x + 2x = 6x$ (like terms, same variable); but $4x + 2y$ cannot be combined (different variables)

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes

- You're planning a concert at the music studio.
- Tickets cost \$12 each, and there's a flat \$50 venue fee no matter how many people attend.
- If t represents the number of tickets sold, how would you write an expression for the total revenue?
- Read each word phrase and sort it into the correct algebraic expression.

Think About It

- What value changes depending on how many people come?
- What stays the same no matter the number of tickets?
- How would you calculate the total if 10 tickets are sold?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Tickets cost \$12 each with a flat \$50 venue fee. If t is the number of tickets, how would you write an expression for total revenue, and which part changes with the crowd?

Level 1 support

Start here: Write $12t + 50$: the $12t$ grows with tickets, the 50 stays fixed.

My expression is ____ because tickets cost \$12 each.

Mi expresión es ____ porque los boletos cuestan \$12 cada uno.

The 50 is a ____ because it ____.

El 50 es una ____ porque ____.

Word bank: Variable Coefficient Constant Algebraic Expression

Listen for: Students write $12t + 50$, identify t as the variable number of tickets, 12 as the coefficient (price per ticket), and 50 as the constant venue fee that does not change.

Level 2

If there were also VIP tickets at \$20 each (v of them), how would your expression change? Generalize how each ticket type adds a new term.

- The new expression is ____ because each VIP ticket adds ____.
- Every new ticket type adds a term because ____.

EXPLORE

How do you turn words into an expression? Why does '12 less than a number g ' become $g - 12$ and not $12 - g$?

Level 1 support

Start here: Key words name the operation, and 'less than' reverses the order.

The words '___' mean ___, so the expression is ___.

Las palabras '___' significan ___, así la expresión es ___.

It is $g - 12$, not $12 - g$, because 'less than' means ___.

Es $g - 12$, no $12 - g$, porque 'menos que' significa ___.

Word bank: Algebraic Expression Variable Coefficient Constant

Listen for: A strong answer matches key words to operations (sum/more than = add, less than = subtract, product = multiply) and explains '12 less than g ' starts with g , giving $g - 12$.

Level 2

Translate 'twice the sum of a number y and 4' and explain why it needs parentheses.

How is it different from 'twice a number y , plus 4'?

- The expression is ___ because 'the sum' must be grouped, so ___.
- Without parentheses, 'twice y plus 4' would be ___, which is different because ___.

CONNECT

A studio charges a \$25 setup fee plus \$8 per recording hour. With h hours, what expression models the cost, and what does each part represent?

Level 1 support

Start here: Write $25 + 8h$: 25 is the fixed setup, $8h$ grows with hours.

The total cost is ____ **+** ____ **h.**

El costo total es ____ + ____ h .

The 25 is the ____ **and $8h$ is the** ____.

El 25 es la ____ y $8h$ es el ____.

Word bank: **Constant** **Variable** **Coefficient** **Algebraic Expression**

Listen for: Students write $25 + 8h$, identify 25 as the constant setup fee and 8 as the coefficient (cost per hour) multiplying the variable h .

Level 2

Another studio charges no setup fee but \$11 per hour: $11h$. For how many hours is the first studio ($25 + 8h$) the better deal? Justify your reasoning.

- The first studio is cheaper when ____ because the higher hourly rate ____.
- Setting $25 + 8h$ equal to $11h$, they cost the same at $h =$ ____, so ____.

Guided Examples

Example 1

Which expression represents 'the product of 6 and a number n'?

Solution: 'Product' means multiplication, so the product of 6 and n is $6n$.

Answer: A. $6n$

Example 2

Which expression represents '9 less than a number y'?

Solution: 'Less than' means subtract from the number, so it's $y - 9$.

Answer: A. $y - 9$

Example 3

Which expression represents 'the sum of a number m and 15'?

Solution: 'Sum' means addition, so the sum of m and 15 is $m + 15$.

Answer: A. $m + 15$

Write About the Math

The Writing Revolution

I can explain my expression using the words variable, coefficient, constant, and algebraic expression.

1. Kernel Sentence subject + verb

Model: Algebraic Expression is a math phrase that has at least one letter.

Expresión algebraica es una frase matemática que tiene al menos una letra.

Write a kernel sentence about algebraic expression. Use a subject and a verb.

Escribe una oración base sobre expresión algebraica. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Algebraic Expression matters in math

Expresión algebraica importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Algebraic Expression matters in math because ____.

Expresión algebraica importa en matemáticas porque ____.

but
pero

Algebraic Expression matters in math, but ____.

Expresión algebraica importa en matemáticas, pero ____.

so
entonces

Algebraic Expression matters in math, so ____.

Expresión algebraica importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about algebraic expression.
Di un hecho verdadero sobre algebraic expression.

Algebraic expression ____.

Question *Pregunta*

Ask a question about algebraic expression.
Haz una pregunta sobre algebraic expression.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about algebraic expression.
Muestra entusiasmo sobre algebraic expression.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with algebraic expression.
Dile a un compañero qué hacer con algebraic expression.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

My expression is ____ **because tickets cost \$12 each.**

Mi expresión es ____ *porque los boletos cuestan \$12 cada uno.*

The 50 is a ____ **because it** ____.

El 50 es una ____ *porque* ____.

The words ' ____ ' mean ____ , **so the expression is** ____.

Las palabras ' ____ ' significan ____ , *así la expresión es* ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. Which expression represents 'the sum of a number m and 15'?

- A. $m + 15$
- B. $m - 15$
- C. $15m$
- D. $m \div 15$

Show your work:

2. Which key word tells you to multiply?

- A. Product
- B. Sum
- C. Difference
- D. Quotient

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that could be modeled by the expression $4n + 10$. Explain what the variable n represents, what the coefficient 4 means, and what the constant 10 represents.

Sentence starter: Situation: _____. The variable n represents _____. The coefficient 4 means _____ for each _____. The constant 10 represents _____.

Show your work:

Reflect — Exit Ticket

Which expression represents '3 more than twice a number n '?

- A. $2n + 3$
- B. $3n + 2$
- C. $2(n + 3)$
- D. $2 + 3n$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $m + 15$ — *'Sum' means addition, so the sum of m and 15 is $m + 15$.*
2. **Try It 2:** A. Product — *'Product' means multiplication. 'Sum' means addition, 'difference' means subtraction, and 'quotient' means division.*
3. **Exit Ticket:** A. $2n + 3$ — *'Twice a number n ' is $2n$. '3 more than' means $+ 3$. So the expression is $2n + 3$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Algebraic Expression is a math phrase that has at least one letter.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).